Requirements from the rabeprazole demo (R10–R13)

For: the software chat. **From:** the owner, after building and running rabeprazole (P3, main 9fc3c5a) on the hosted demo, 25–28 Oct 2026. **Goes with:** `RAB26002-batch-record.pdf`, the model for R11. This document replaces the separate R11 PDF of 26 Oct.

#	Subject	In one line
R10	Bugs found in the live batches	20 defects; the potency calculation that shows no numbers is the first
R11	The whole batch record together, on one screen	One page per batch with everything from the ERP order to the release: every stage, step, value, person, time and signature; view only, no download
R12	Dissolution as an in-process test	A block that records dissolution during manufacture, not only as a lab release test
R13	See and change the potency before dispensing	Show the lot's potency and the calculated weights before weighing; a clear QC path to change the potency %

Priority marks: **P1** wrong or blocks a batch · **P2** misleading on screen · **P3** wording or polish. Each fix follows `fix-finding` (reproduce first, a test, database guard where it is a control).

R10 · Bugs found in the live batches

Seen on RAB26001, RAB26002 (released) and RAB26003 (owner's own run, 25 Oct). Step numbers are the rabeprazole MBR's (RAB-WG-DEMO v1 r2).

#	Pri.	What happens	What it should do
1	P1	**Potency calculation shows no numbers.** After "Calculate from the lots" (step 1.2.1, TB-01) the screen shows only "Calculated from the lots (see the entries)", an entry reading "potency calculation", and two computed boxes with no value. The saved entry holds as-is 98.8036 %, active 3.037 kg, MCC 4.464 kg, but the operator never sees them.	Show the result in the Potency calculation box and in the entry: lot, assay, water, basis, as-is potency, the active's target, the compensating excipient's target, the total. Show each target as the computed box's value before weighing.
2	P1	**Weighing panel compares with the order, not the potency target.** At 1.2.1 "Weigh one container" says "still to weigh 3.000 kg" and "4.500 kg"; after the correct weights it says "0.037 kg over the order" and "0.036 kg still to weigh".	For a potency-adjusted line, compare with the calculated targets (3.037 / 4.464 kg) and say so ("target from the potency calculation").
3	P1	**No recovery after Run end.** If the sealing step's recipe check (1.3.3, TB-21) is not recorded before "Run end", the step can never be signed and the packaging order can never complete (RAB26001/P01 is stuck for ever). There is no void, abort or reopen.	A controlled way out: QA reopens the run, or voids and repeats the step with a reason, signature and linked deviation, all in the record. Better still, warn before "Run end" while a required check is missing.
4	P1	**Bulk issue offers the wrong containers.** "Containers of batch RAB26001" offers every closed container for packaging: granules, cores, coated tablets and the inspection-rejects container that was destroyed at 6.2.2.	Offer only the released bulk (the final intermediate of the MBR). Never offer destroyed or reject containers.

5	P1	**The processing record is never locked.** /records/RAB26002 shows "Unlocked, not locked yet" after QA review and release.	Lock it at the end of review (or at release, whichever the plan says), as the packaging record is.
6	P2	**A released lot still shows the release form.** Release cockpit → RAB26002/P02 (released) still offers the decision form.	Show "Released by … on …" and no form.
7	P2	**Stale disposition button.** The step card still showed "Disposition the exception (QA)" for EX-1 after it was dispositioned; Sign then refused (DI-26). The run overview also shows stale "N open" marks for a few seconds.	Refresh the card and the marks after each act.
8	P2	**QA act offered to an operator.** Arjun (operator) sees "Disposition the exception (QA)…" as the primary action.	Show it only to roles that may do it; to others, show who can.
9	P2	**Wrong product name.** The active's quantity is labelled "Exemplazine sodium dispensed (potency-adjusted)" on a rabeprazole batch (step 1.2.1, the exceptions, the record).	Use the material name from the MBR or BOM.
10	P2	**"Test" wording on GMP.** The lot hint says " (TEST- lots only)". A refusal says "recorded in the test run" on a GMP batch.	No sandbox wording on the GMP installation.
11	P2	**Yield shown unrounded.** "99.132233779667835181886098500086 2 %" (granulation, 2.2.7). The limit line reads "pass low 99.1 high 99.1". The validated range cites another product's report (VR-RTW).	Show the value at its declared precision, show a sensible limit text, and cite the product's own report.
12	P2	**The finished lot box gives nothing back.** After "Post the finished lot in quarantine" it shows only "Saved." with the fields emptied.	Show the lot code with a link to it.
13	P2	**No link to the containers.** The run and step pages have no link to the output containers; the only path is through Complaint lookup → trace forward. Line completion has no link from the packaging run.	Add the links from the run, the step and the packaging run.
14	P3	**Internal codes on screen:** "dispMcc at p-disp-api@", "step p-g21@ is not performed", "granulator.power_kW", "EXECUTION_COMPLETE; 7 usable containers".	MBR names, step numbers and plain states everywhere.
15	P3	**Unit placeholders:** "1000 {tbl}" (planning card); printed lots show the unit as "1" ("20000 1").	"tablets", "labels", "units".
16	P3	**The finished lot's page offers starting-material forms** (assay basis, artwork version, identity test).	Offer only the forms that fit a finished lot.
17	P3	**Anti-tampering check on an India-only order.** Cartoning offered "Record: Atd check" on an IN-only order.	Offer it only when a market of the order needs it, or say why it is asked.
18	P3	**Misleading message while loading:** "No open batch to reserve for." flashes before the batches appear (lot reservation).	Show "Loading…" until the list is in.

19	P3	**Packaging order numbers skip.** The second batch's packing order is RAB26002/P02, not /P01 (the {SEQ} counts across parent batches).	Owner's default: number per parent batch (/P01 for each batch's first order), unless the numbering rule says otherwise.
20	P3	**Entry summaries are only the kind's name** ("area occupancy", "end point", "recipe check", "count": about 60 kinds).	A plain sentence with the values, per kind (also needed by R11).

R11 · The whole batch record together, on one screen

Model: the attached `RAB26002-batch-record.pdf` (20 pages), made from the hosted demo's own data for the released batch RAB26002 and its packing order RAB26002/P02.

> **Owner's request:** "this pdf is what i was talking about, we want this like in the software. The PDF should not have a download option, but with this method we should be able to view it."

1. The gap today

Where in the software	What it gives	Problem
Shop floor → run → each step	Every step's values, entries and signatures	One step at a time; 63 clicks for one batch
Batch record review	Review acts, exception register, containers	No stage or step content
Archive → "PDF/A-3 true copy"	Everything (704 pages for RAB26002)	Raw table rows and JSON, not readable by a person

The plan already expects a **human-readable** rendering: the outline's archival section names a "canonical PDF/A-3 rendering", and ADR 0010 rejected frozen JSON alone because it is "not a human-readable true copy". There is no page today that reads like a batch manufacturing record.

2. What to build

One **Batch record** page per batch that tells the **whole story, from the ERP order to the release** (owner, 28 Oct: "from ERP order to release, all details should be there"). It covers the processing record and its packaging order(s) together, on one scrolling page, in the order things happened. Every item shows who did it, when, and the signature with its meaning.

1. **Header:**
   - product, strength, batch number, batch size;
   - MBR family, version and revision;
   - markets, made and expiry dates;
   - status, review state, release state;
   - the record's content hash.
2. **ERP order:** order number and ERP reference, product, quantity, the order's BOM, when it was received, and the BOM-versus-MBR check.
3. **Planning and batch issue:** the batch number allocated (rule and serial), who planned and issued it, the MBR pinned, and when.
4. **Materials:**
   - every lot reserved: lot, material, quantity, its status and expiry, who reserved it;
   - for an API, its assay, water and as-is potency at the time;
   - later, what was actually consumed from each lot.

5. **Before production:** the cleaning of each room and machine (who cleaned, who checked, the cleaning status at use), the room clearances and the equipment booked.
6. **Manufacturing:** stages at a glance, then each stage and each step, with:
    - status;
    - room and equipment;
    - the **recorded values** (value, unit, how captured, who, when, any correction and its reason);
    - the **entries** in plain language, with the in-process tests in their steps (disintegration, and R12's dissolution);
    - the **signatures** (performed, verified, witnessed);
    - the **exceptions** and their dispositions;
    - and the stage sign-off.
7. **Containers and yields:** every container (OC-…) with net quantity, who closed it, who verified it, its state and where it went; the yields and the reconciliations of each stage.
8. **Packaging:** each packaging order of the batch: its ERP order, the bulk issued, line clearance, then every packaging step. It also shows:
    - printed materials issued, the coding specimens (start, middle, end) and their reconciliation;
    - destruction of surplus;
    - packs, shippers, packaging yield;
    - the finished lot posted in quarantine.
9. **Laboratory:**
    - samples taken;
    - every release test of the specification with its points (e.g. dissolution acid 2 h and buffer 45 min): value, limits, verdict, analyst, approver;
    - any OOS and how it closed;
    - the reserve sample.

    These release results are shown here as part of the release, as on the release cockpit. They are not added to the manufacturing steps.
10. **Exceptions and deviations:** the full register, with each floor disposition and QA review outcome, and the linked deviation, CAPA or change records.
11. **Review:** production approval, audit-trail review, QA review (each signature, with any reopening and its reason).
12. **Release:**
    - the release checklist as it stood when signed;
    - the decision, the authorised person and the time;
    - the markets;
    - anything after release (suspension, recall, amendment).
13. **Audit trail:** its own section at the end (collapsed by default), or a link to the existing audit-trail page.

The model PDF (`RAB26002-batch-record.pdf`) shows items 1, 6, 7, 10 and 11, and the packaging steps of item 8. It does not yet show items 2–5, 9, 12 and 13, or the printed-material detail of item 8. The view must add them in the same style.

Where to reach it: a "Batch record" link on these pages:

- the Shop floor run page;
- Batch record review;
- Batches;
- Packaging orders;
- Release cockpit;
- Archive;

- Complaint lookup / genealogy.

3. View only: no download

- It is a **screen view**: no download, export or "save as PDF" button on it.
- **Printing is turned off for this page**: a print stylesheet that shows only "Printing is not available; use the Archive for a controlled true copy". Screenshots cannot be prevented, so every screen carries a **watermark** with the viewer's name and the time, plus "Controlled view, not a copy".
- Reading it needs a record-view permission held by the review, QA, production-head and QC roles. Floor roles see only their own batch, and a contract giver only its loan-licence batches.
- Opening the record is written to the access log (who viewed which record and when).
- A batch that is still running shows the record so far, with a red **"IN PROGRESS: not final"** band. A locked record shows "Final, locked on …".

> **Keep the controlled copy for inspectors.** The regulations require a site to provide accurate and complete copies of records, both human-readable and electronic, for inspection. The **Archive** true-copy export therefore stays, but only for the archive and QA roles, with a reason and an audit entry. It should use the **same readable layout** as this view, instead of the raw rows it has today (the record's data stays embedded, as now). Only the everyday view has no download.

4. Text fixes the view needs (seen while making the model)

Seen in the demo data	Wanted on the record
About 60 entry kinds summarise as their own name only ("area occupancy", "end point", "recipe check", "count")	A plain sentence per kind with its values, e.g. "End point by time: 31.6 s, within range", "Recipe RCP-CF-RTW v1 matches"
The potency calculation summarises as "potency calculation"	"Lot RAB-API-0001: assay 99.40 %, water 0.60 % → 98.8036 % as is; active 3.037 kg; MCC 4.464 kg" (this also fixes the weighing screen, which shows no targets after Calculate)
Units `{tbl}`; yield `100.004943429…` %	"tablets"; a yield at its declared precision
Internal tags (`blister.seal_temp_C`), codes (`MANUAL_ENTRY_FALLBACK`), item keys (`dispApi`), ids	The MBR names ("Sealing temperature, °C"), plain exception names, never ids
The active labelled "Exemplazine sodium" on a rabeprazole batch	The material name from the MBR or BOM

5. How the model was made (for the build)

Everything in the model came from views the server already serves:

- `/api/batches/{id}` (stages, sign-offs, exceptions);
- `/api/batches/{id}/steps/{stepKey}` for each of the 52 + 11 steps (fields with attempts, entries with data, signatures, exceptions);
- `/api/review/batches/{id}` and `…/register`;
- `/api/containers`.

The sample generator is a 200-line script. In the product:

- One server view should assemble the record from the stored record content as of a point in time, and show its content hash, so the page and the true copy say the same thing.
- The entry texts belong in the block definitions next to each entry schema.

6. Done when

- The page exists for processing and packaging records.
- For RAB26002 and RAB26002/P02 it shows all 63 steps, matching the step views value for value, with no raw codes, ids or JSON.
- There is no download or print control; the watermark shows; viewing is permission-checked and logged.
- The Archive true copy uses the same readable layout and stays restricted.
- Covered by tests:
    - a browser test opens the record from each place listed in section 2 and checks that there is no download control;
    - the golden batches compare the rendered record;
    - tests carry the IDs of the controls the slice touches (copies, access, audit trail).

Open owner question (the default is used unless the owner says otherwise): should production supervisors on the floor see the full record? **Default:** yes, for batches of their own area.

R12 · Dissolution as an in-process test

Today: dissolution exists only as a lab release test on the finished lot (specification test, two-stage profile, slice 7p). No MBR block records dissolution during manufacture. Disintegration is an in-process block (rabeprazole step 5.2.11). **Owner:** dissolution should also be possible as an in-process test.

Why plants need it:

- the acid-resistance check of enteric-coated cores before they are released to packing;
- a dissolution check of sustained-release cores before coating;
- a check after a coating-level change or a reprocessing step.

The result decides whether the batch goes on, so it must be in the batch record.

What to build: a new tablet block, "In-process dissolution", that can be placed in any stage of an MBR.

1. **In the MBR:**
    - apparatus (basket or paddle), medium, volume, temperature, speed;
    - the time points (one or several; two-stage acid → buffer for delayed release);
    - units per stage (6);
    - limits per point;
    - the staging rule of the pharmacopoeia cited (for example S1/S2/S3, or A1–A3 and B1–B3 for delayed release, with the edition: IN-33);
    - sample size;
    - whether the batch waits for the result (hold point) or goes on at risk.
2. **Two ways to record, chosen in the MBR:**
    - **(a) on the floor or in the IPC lab:** the analyst records the start (instrument ID, calibration status checked at point of use), each unit's % released at each point, and the end. The block computes the mean and the stage outcome. When a stage fails, it asks for the next stage's units.
    - **(b) as a sample to QC:** the step sends an in-process sample request (to LIMS-lite or the LIMS), and the result comes back into the step. A hold point keeps the next stage closed until the approved result is in.
3. **Instrument data:** the values can come from a dissolution tester or its software (adapter), or be typed with the manual-entry rules (second person, exception), the same as for balances.
4. **Results:**
    - a failing result raises an exception and blocks the stage sign-off until it is dispositioned or a deviation is linked;

    - a result outside the alert limits is flagged;
    - the result shows in the step, the stage summary, the R11 record view and the review.
    - **Where results show (owner, 28 Oct):** an **in-process** dissolution result belongs in the batch record (the step, the stage summary, the R11 record view and the review). A **release** dissolution test stays where it is today, on the finished lot and in the release cockpit. Nothing more is needed for release tests.
5. **Demo:** add it to the rabeprazole MBR as an acid-resistance check of the coated tablets (2 h in 0.1 M HCl, at most 10 % released), before inspection, and to the guide's golden batch.

R13 · See and change the potency before dispensing

Owner's questions (25 Oct): "Can I see the potency calculation before dispensing?" and "I want to adjust the potency percentage." Today:

- the lot's potency (assay 99.40 %, water 0.60 %, as-is 98.80 %) and the calculated weights show nowhere before the weighing step;
- even there, the weights do not show (R10 #1);
- changing the potency % is possible: QC records a new ASSAY result, the head of QC approves it, QC records the lot assay with a reason, all before the operator presses Calculate. But the path is spread over several pages and no guide or screen explains it.

What to build:

1. **Potency on the lot:** Material lots → an API lot shows its current assay, water, basis, as-is potency, the results it came from, and its history of changes (who, when, why).
2. **Preview before dispensing:** at lot reservation and on the batch's planning page, a "Potency preview" shows what TB-01 will calculate for the reserved lot(s): the active's target and the compensating excipient's target. It is marked "preview: the weighing step recalculates" and is never a recorded value.
3. **Change the potency %, one guided page for QC:**
    - "Update the potency of lot X": choose the new approved ASSAY and WATER results, give a reason, sign;
    - the preview updates;
    - if the batch has already calculated, the step shows "Potency changed since the calculation: recalculate", and the change is an exception for QA;
    - typed values stay refused (results only), as now.
4. **Never** let the floor type a potency % or a target weight. The controls stay as they are (PR-19, DI-33).

> **Still waiting on the owner (not for building yet):** R9 "Demo observer", a view-only login for an audience to watch a live batch, was offered on 23 Oct and not confirmed.